Customer Outage Request (Illustrative Case Study)

Author: Oleksiy Milner · linkedin.com/in/oleksiy-milner-240a3969

An end-to-end system analysis of a common field service scenario, modeled as a business process, a data model and an API contract. All names and data are illustrative; no real company data is used.

Scenario

A customer reports an internet outage. The provider verifies the contract and SLA, creates work in a team queue (bucket), finds a technician by skill, zone and availability, and reassigns the work. The technician completes the repair, or schedules a follow-up visit if a spare part is missing. The customer confirms, and the activity is completed.

Actors: Customer, Dispatcher, Technician, DEMO Service (system)

Deliverables

Artifact	What it shows	Tool
BPMN process	Roles, SLA decision, escalation, customer confirmation	Camunda Modeler
Conceptual ERD	Core entities and relationships	draw.io
OpenAPI 3.0 spec	Create and update activities, JWT security	Swagger Editor

Key design decisions

- **SLA check:** an exclusive gateway with a separate reject path that informs the customer.
- **Escalation:** a 30-minute timer boundary event routes unmatched work to the dispatcher for manual assignment, instead of rejecting a valid request.
- **Follow-up loop:** a missing spare part triggers a follow-up visit, then returns to the repair step.
- **Bucket, then reassign:** work is created in a team queue and reassigned to a qualified technician. The API returns `422` if the technician lacks the required work skill.
- **Data model:** a resource can be a technician or a bucket; routes are created automatically; activity-skill and resource-skill are many-to-many relationships.
- **API contract:** JWT bearer security, standard `400`/`401`/`404` responses, `readOnly` server-generated fields, and a separate update schema for `PATCH`.

Traceability

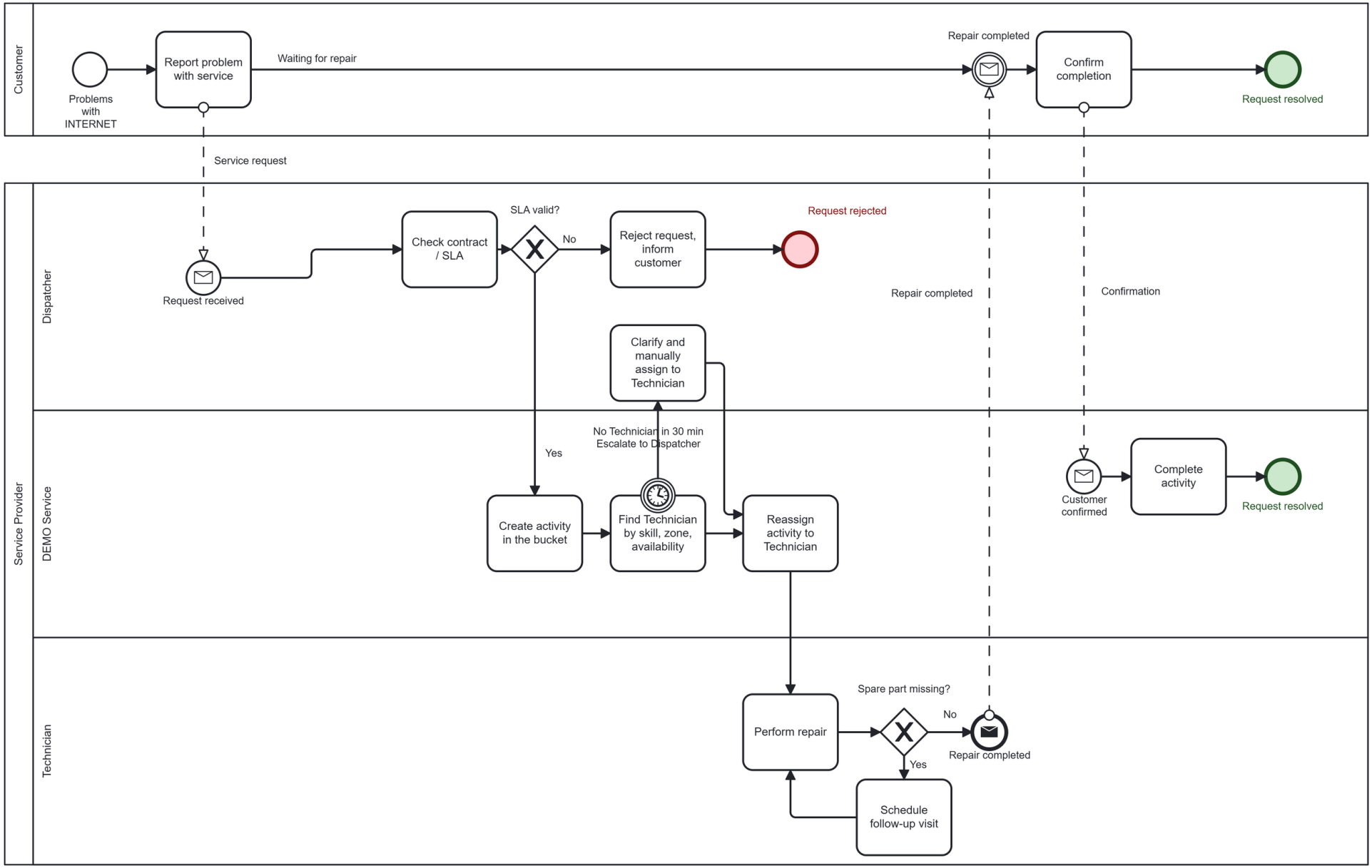
BPMN step	API	Data entity
Check contract / SLA	internal business rule (out of API scope)	Service Request
Create activity in the bucket	`POST /activities`	Activity, Resource
Find technician by skill, zone, availability	internal matching (out of API scope)	Resource, Work Skill
Clarify and manually assign / Reassign activity	`PATCH /activities/{id}` (`resourceId`; `422` on skill mismatch)	Activity, Resource, Work Skill
Perform repair / follow-up visit	`PATCH /activities/{id}` (`status`)	Activity
Complete activity	`PATCH /activities/{id}` (`status: completed`)	Activity

Assumptions and scope

- The SLA check and technician matching are internal business rules and are not modeled in the API demo.
- After escalation, the dispatcher resolves the request manually, outside this model.
- `resourceId` on activity creation is the target bucket or technician; the route is created automatically.

Business process (BPMN)

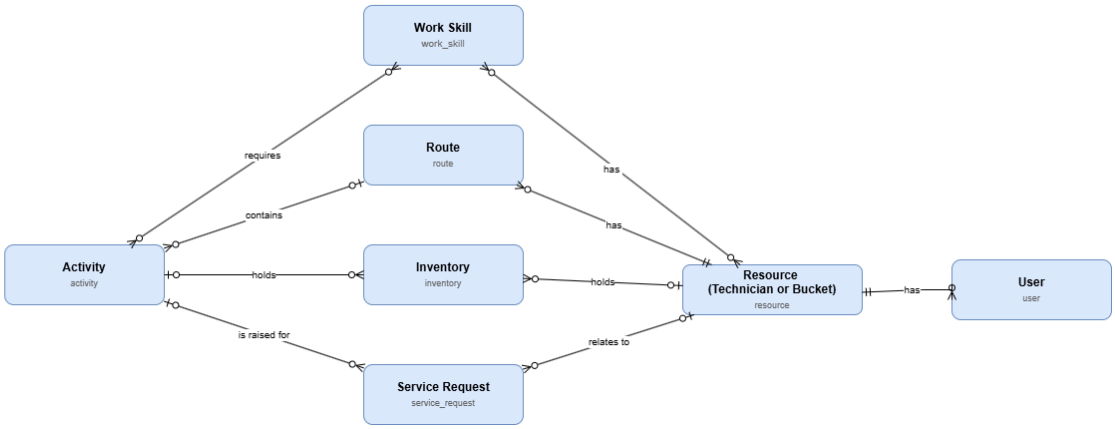
Customer and service provider pools with dispatcher, system and technician lanes; SLA gateway, 30-minute escalation timer and message flows for confirmation.



Data model (ERD)

Conceptual view of the core entities. A resource is either a technician or a bucket.

DEMO: conceptual ERD (core entities)



API contract (OpenAPI 3.0)

Endpoints: `POST /activities`, `GET /activities/{activityId}`, `PATCH /activities/{activityId}`. *Rendered in Swagger Editor.*

